



Lab 03: Explaining Intelligent Agents – Research, Understand, and Share (Blog-Style)

Objective: Research a focused subtopic from *Intelligent Agents* (Lecture 2) using academic sources. Make sense of the literature using scholarly discovery and comprehension tools. Produce a clear, original, **first sketch** of an informative blog post (public link) by the end of the lab. Practice technical communication and proper referencing (APA or IEEE).

Introduction

This activity builds on Lecture 2 (Intelligent Agents). You will select one focused subtopic — such as the PEAS framework, environment types, agent architectures, or rationality — investigate it through academic sources, and communicate what you have learned as a short, well-referenced blog post published on a public link.

Group Size: Work in groups of **3** with those seated closest to you. Do not move around. Members should be from the same row or immediate vicinity.

Lab Activity

Section 1: Deliverables (Due by end of lab)

1. **First Sketch (Draft) Blog Post:** Publish a shareable link (Medium, Notion, Google Sites, or similar).
 - Length target for draft: **~600–900 words** (final version can be 800–1200).
 - Include at least one visual (diagram/table) and a provisional references section.

2. **References List:** 4–5 academic sources (research papers/textbooks/reputable labs/universities).
3. **Submission:** One link per group via Moodle (add group members' names & IDs).

Section 2: Approved Research & Comprehension Tools

Google Scholar

1. Search key phrases (e.g., *PEAS intelligent agents*, *utility-based agents*).
2. Use the **“Cite”** feature for APA/IEEE; open publisher or preprint links.
3. Filter by year for recent surveys/tutorials.

Connected Papers

1. Start from a known seed paper or textbook chapter title.
2. Explore prior/derivative works; identify review/survey papers for faster grounding.
3. Export a reading list; follow links to PDFs where available.

Research Rabbit

1. Create a collection for your topic; add 2–3 seed papers.
2. Use *Similar Work* and *Co-cited* graphs to find clusters.
3. Bookmark candidate sources to decide your final 4–5 references.

NotebookLM (for understanding only)

1. Upload PDFs/links; ask clarifying questions to build your *own* notes.
2. Extract definitions, comparisons, and key arguments for your outline.
3. **Do not copy text;** use it to understand, then write in your own words.

Academic Integrity & Tool Policy

Allowed: Use discovery/comprehension tools (Google Scholar, Connected Papers, Research Rabbit, NotebookLM) to *find and understand* sources.

Not Allowed: Any use of LLMs/AI text generators (e.g., ChatGPT, Gemini, Copilot) to **write**

or **paraphrase** your blog content.

Write the post yourselves. Copy-paste or AI-written text ⇒ **zero** for this practical.

Section 3: Suggested Topics (choose one)

- **PEAS Framework:** Why it matters; build a PEAS table for a real system.
- **Environment Types:** Fully vs. partially observable; deterministic vs. stochastic; episodic vs. sequential; static vs. dynamic; discrete vs. continuous; single vs. multiagent – with examples.
- **Agent Types & Architectures:** Simple reflex, model-based, goal-based, utility-based, learning agents – compare with a use-case table.
- **Rationality in Agents:** What makes an agent rational? Performance measures, uncertainty, trade-offs.
- **Mini Case Study:** Design analysis for one application (e.g., robotic vacuum, self-driving, stock bot) using PEAS + environment classification + agent type justification.

Your chosen topic and angle

PEAS Framework: Understanding a Game-Playing Intelligent Agent

Section 4: Recommended Blog Structure

1. **Title** – Clear and specific.
2. **Introduction** – What the concept is and why it matters.
3. **Core Explanation** – Definitions and key ideas (in your own words).
4. **Deep Dive** – Compare/contrast (e.g., agent types or environment dimensions) with a small table.
5. **Real-World Example** – Apply the concept to a concrete system.
6. **Common Pitfalls** – Misconceptions or edge cases.
7. **Conclusion** – Takeaways and implications.
8. **References** – 4–5 academic sources (APA or IEEE).

Visuals to Include (at least one)

- A **PEAS** table for your chosen system.

- An **environment classification** checklist/table.
- A small **agent type comparison** table.

Section 5: Where to Publish (pick one)

Medium (fast & polished)

1. Create an account → *Write a story*.
2. Paste your content; add headings, images, tables (as images if needed).
3. Publish → set visibility to **Public** → copy link.

Notion (collab & structured)

1. Create a page → write with headings, callouts, tables.
2. Insert image/table for PEAS and comparisons.
3. Share → **Publish to web** → copy link.

Google Sites (simple web page)

1. Create a new site → add text/sections.
2. Insert images or screenshots of tables.
3. Publish → copy public link.

In-Lab Timeline (2 hours)

Time Box	Task	Output
0–20 min	Pick topic, define angle, assign roles (research, visuals, drafting)	One-line topic + outline bullets
20–65 min	Find sources (Scholar / Connected Papers / Research Rabbit), read abstracts, skim key sections	Short notes + 4–5 shortlisted references

Time Box	Task	Output
65–95 min	Draft first sketch (intro, core explanation, one visual, provisional references)	600–900 words draft
95–115 min	Group review (clarity, flow, citations), polish visual(s)	Revised draft
115–120 min	Publish & submit link on Moodle	Public URL submitted

Referencing

- Use **APA** or **IEEE** style consistently.
- Cite images/diagrams you did not create.
- Prefer textbooks, survey/tutorial papers, and reputable lab/university pages.

Assessment (20%)

Criteria	Details	Marks
Quality of Sources	4–5 academic references; relevance; diversity (textbook/survey/tutorial + recent work)	5
Accuracy & Depth	Correct definitions; nuanced analysis; appropriate examples	5
Clarity & Structure	Readable, logically organized, engaging for a 2nd-year audience	4
Visuals	At least one meaningful table/diagram (PEAS, environment, agent comparison)	3
Referencing	Consistent APA/IEEE; proper attributions	3

Integrity Declaration

By submitting your link, you affirm the post is written entirely by your group members without AI text generation or paraphrasing tools. Any violation will result in a zero for this practical.

Submission Instructions

1. Publish your draft (Public link).
2. Submit on Moodle:
 - o Post Title
 - o Public URL
 - o Group members' names & IDs
 - o Reference list (copy into the submission box as well)

Submission Details

Post title and public URL

PEAS Framework: Understanding a Game-Playing Intelligent Agent

https://app.notion.com/p/PEAS-Framework-Understanding-a-Game-Playing-Intelligent-Agent-3b877015578380d1bbaefa29ed10297e?source=copy_link

Group members' names & IDs

Somarathne H.D.P.Y - IT24101854

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Lalanga J.A.L - IT24103581

Reference list (4–5 academic sources, APA or IEEE)

[1] S. Franklin and A. Graesser, "Is it an agent, or just a program?: A taxonomy for autonomous agents," in *Intelligent Agents III: Agent Theories, Architectures, and Languages*, pp. 21–35, Springer, 1997.

[2] J. Schaeffer, N. Burch, Y. Björnsson, A. Kishimoto, M. Müller, R. Lake, P. Lu, and S. Sutphen, "Checkers is solved," *Science*, vol. 317, no. 5844, pp. 1518–1522, 2007, doi: 10.1126/science.1144079

[3] S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2020.

[4] P. Stone and R. S. Sutton, "Scaling reinforcement learning toward RoboCup soccer," in *Proc. 18th Int. Conf. Machine Learning (ICML)*, pp. 537–544, Morgan Kaufmann, 2001.

Quick Checklist (before submitting)

- 600–900 words (first sketch) with clear intro & core explanation
- At least one visual (PEAS / environment / agent comparison)
- 4–5 academic references listed (APA/IEEE)
- Public link works (incognito test)
- No AI-generated text; written by us

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FACULTY OF COMPUTING